
Toscanini 05 transfer manifest

invbench-p03-phe-biofouling156ae5155c73 · run r401-bench-p03-phe-biofouling · maturity
EARLY_TECHNICAL_EVALUATION · 3D class CONCEPTUAL_3D

What is transferable

the recorded invention architecture for inv:bench-p03-phe-biofouling:156ae5155c73 with its evidence roles, the engineering projection (design inputs, failure modes, verification items, build plan, equations), the unknown roadmap, and the 3D artifacts this class earned (CONCEPTUAL_3D)

Artifact inventory

every file with sha256 in PACKAGE_MANIFEST.json; the machine-readable layers listed in 00; MODEL/ carries the parametric source and derived CAD when ENGINEERING_3D is earned

Engineering maturity

EARLY_TECHNICAL_EVALUATION — basis and counts in MATURITY_BASIS.json

Buyer capability needed

an engineering team able to take ENGINEERING_PROPOSED state to manufacturable definition (the design work named in the unknown roadmap) and to run the pre-registered decisive experiment — the context-specific buyer capabilities are recorded in the buyer decision card (What must the buyer bring?), derived from the canonical applicability state

You must verify

the buyer carries the verification duty for every transferred claim (nothing arrives pre-verified):

- VF-001: Accelerated aging test with simultaneous chloride exposure and mechanical stress while monitoring healing agent depletion and repair effectiveness
- VF-002: run killer experiment: falsification_test_from_candidate (EIG/cost 0.4557178723356231)
- VF-003: temperature-rise mapping at max duty
- VF-004: temperature-rise mapping at maximum duty with IR or embedded sensing (measured quantity: temperature rise)
- VF-005: thermal-resistance trend over accelerated cycling (measured quantity: mechanical stress / fatigue endurance)
- VF-006: temperature-rise mapping at max duty
- VF-007: worst-case ambient soak with staged power steps and stability monitoring (measured quantity: power / energy deposition)

Buyer and regulatory requirements (canonical applicability)

canonical problem context: INDUSTRIAL_PROCESS

buyer type: industrial process-equipment / component manufacturer (candidate buyer class for an industrial process technology) [APPLICABLE]

engineering capability: process/mechanical engineering with the domain's physics (see the engineering specification's governing model) [APPLICABLE]

engineering capability: field/plant integration engineering for the recorded operating environment [APPLICABLE]

manufacturing capability: general industrial fabrication (machining, welding/brazing, sheet/plate work, assembly) — the specific processes are the domain candidates recorded in the engin... [APPLICABLE]

manufacturing capability: quality control appropriate to the recorded failure modes [APPLICABLE]
regulatory: applicable industrial codes/standards (e.g. pressure equipment, piping, electrical)
UNKNOWN until determined for the installation jurisdiction — the buyer's regulatory affairs makes the determination; medical-device regulation is NOT_APPLICABLE... [UNKNOWN]
medical regulatory: medical-device regulatory requirements (FDA 510(k)/PMA, ISO 10993
biocompatibility, ISO 13485 medical QMS) [NOT_APPLICABLE]
market channels: industrial process / water-utility / plant engineering channels appropriate to the recorded application [APPLICABLE]

Required tooling

CAD (STEP readers; CadQuery/OCCT for parametric rebuilds — see MODEL/PARAMETRIC_MODEL_SOURCE.py), mesh inspection (any STL viewer; trimesh used for the independent watertight check), and the bench equipment named in the build plan

Dependencies

no external services; the package is self-contained with real hashes (verify after transfer against PACKAGE_MANIFEST.json)

Confidentiality classification

CONFIDENTIAL — technical evaluation draft for the recipient's engineering diligence only; not a release document, not legal advice

Reproduction instructions

rebuild the geometry from MODEL/PARAMETRIC_MODEL_SOURCE.py + MODEL/PARAMETERS.json (the regeneration check records the expected measurements); verify every file against the manifest hashes

Open technical work

7 unknowns with resolution actions in UNKNOWN_ROADMAP.json — this is the recipient's first work queue

Epistemic boundary: DESIGNED / SIMULATED / EVIDENCE-SUPPORTED / EXPERIMENTALLY VERIFIED states come only from the canonical invention object. EXPERIMENTALLY VERIFIED requires real reality-loop evidence. Potential IP territory — formal legal review required. Toscanini performs technology discovery; patentability, claim drafting, infringement/FTO and legal opinions are matters for qualified IP counsel.